Claudia Rella

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Institut des Hautes Études Scientifiques (IHES)
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RESEARCH PROFILE

I am a mathematical physicist working at the interface of algebraic geometry, number theory, and topology with quantum field theory and string theory. My work explores the interplay between resurgence, arithmetic, and quantum modularity, with applications to non-perturbative phenomena in topological string theory and quantum topology.

EMPLOYMENT

2024 – present **Postdoctoral Researcher in Mathematical Physics**
Institut des Hautes Études Scientifiques (IHES), France
Mentored by **Prof. Maxim Kontsevich** and funded by the **Huawei Young Talents Program**

EDUCATION

2020 – 2024 PhD in Mathematical Physics — *Très Bien (highest honours)*
Department of Theoretical Physics, **University of Geneva**, Switzerland
Thesis: *On the arithmetic of resurgent topological strings*
Supervised by **Prof. Marcos Mariño** and funded by the **ERC SyG ReNewQuantum**

2018 – 2019 MSc in Mathematical and Theoretical Physics — *Distinction (highest honours)*
Mathematical Institute and Department of Physics, **University of Oxford**, UK
Thesis: *An introduction to motivic amplitudes*
Supervised by **Prof. Francis Brown** and funded by the **Torno Subito Scholarship**

2015 – 2018 BSc in Physics — *Summa cum Laude (highest honours)*
Department of Physics, **Sapienza University of Rome**, Italy

PUBLICATIONS

IN PREPARATION

(with V. Fantini) *Modular resurgent structures 2*

(with M. C. N. Cheng, I. Coman, and V. Fantini) *Vector-valued modular resurgence*

PREPRINTS

(with V. Fantini) *Modular resurgent structures*, submitted (under review, 2025) [arXiv:2404.11550]

REFEREED RESEARCH ARTICLES

(with V. Fantini) *Modular resurgence, q-Pochhammer symbols, and quantum operators from mirror curves*, **Lett. Math. Phys.** 116, 39 (2026), https://doi.org/10.1007/s11005-026-02063-x [arXiv:2506.08265]

(with V. Fantini) *Strong-weak symmetry and quantum modularity of resurgent topological strings on local* $\mathbb{P}^2$, **Commun. Number Theory Phys.** 19 (2025), no. 1, pp. 1-73, https://dx.doi.org/10.4310/CNTP.250215002033 [arXiv:2404.10695]

(with M. Mariño) *On the structure of wave functions in complex Chern–Simons theory*, **SIGMA** 22 (2026), 002, 45 pages, https://doi.org/10.3842/SIGMA.2026.002 [arXiv:2312.00624]

Resurgence, Stokes constants, and arithmetic functions in topological string theory, **Commun. Number Theory Phys.** 17 (2023), no. 3, pp. 709-820, https://dx.doi.org/10.4310/CNTP.2023.v17.n3.a4 [arXiv:2212.10606]

(with B. Döbrich and T.-T. Yu) *Searching for muonphilic dark sectors with proton beams*, **Phys. Rev. D** 106 (2022) 3, 035023, https://doi.org/10.1103/PhysRevD.106.035023 [arXiv:2205.09870]

An introduction to motivic Feynman integrals, **SIGMA** 17 (2021), 032, 56 pages, https://doi.org/10.3842/SIGMA.2021.032 [arXiv:2009.00426] (with A. Frankenthal et al.) *Characterization and performance of PADME's Cherenkov-based small-angle calorimeter*, **Nucl. Instrum. Methods Phys. Res. A** 919 (2019) 89-97, https://doi.org/10.1016/j.nima.2018.12.035 [arXiv:1809.10840]

PROCEEDINGS

(with M. Alim, V. Fantini, and L. Hollands) *Mini-Workshop: Resurgence, Difference Equations and Quantum Modularity*, **Oberwolfach Rep.** 22 (2025), no. 4, pp. 2593-2632 (main reporter), https://doi.org/10.4171/owr/2025/48

DOCTORAL THESIS

On the arithmetic of resurgent topological strings, Doctoral Thesis Sc. 5852, **University of Geneva** (2024), https://dx.doi.org/10.13097/archive-ouverte/unige:181979

TALKS AND SEMINARS

2026, May	Mathematical Physics Seminar, LPTHE, Sorbonne Université, France
Mar	Workshop on Mathematical Structures of Amplitudes, ESI, Austria
2025, Nov	Asymptotics, Geometry, and Physics Seminar, University of Southampton, UK
Oct	Mini-Workshop on Resurgence, Difference Equations, and Quantum Modularity, MFO, Germany
Aug	ReNewQuantum Closing Conference, SDU, Denmark
May	Random Matrix Theory Seminar, University of Oxford, UK
May	Mathematical Physics and Algebraic Geometry Seminar, CMSA, Harvard University, USA
Apr	Conference on Higher Structures, Moduli Spaces and Integrability, Universität Hamburg, Germany
Mar	Workshop on The Arithmetic of Calabi-Yau Manifolds, MITP, Germany
Jan	Youngst@rs Physics and Number Theory, MITP, Germany
2024, Nov	The Seed Seminar of Mathematics and Physics, IHP, France
Oct	Workshop on Holonomic Techniques for Feynman Integrals, MPP, Germany
Sep	GAP XIX: Moduli Spaces and Higher Structures, Sapienza University of Rome, Italy
May	String Theory Seminar, University of Geneva, Switzerland
May	Workshop on Resurgence and Modularity in QFT and String Theory, GGI, Italy
May	Mathematics Seminar, Yale University, USA
Apr	String Theory Seminar, CERN, Switzerland
Mar	Seminar on Quantum Modularity and Resurgence, IHES, France
Feb	Workshop on Positive Geometry in Particle Physics and Cosmology, MPI MiS, Germany
2023, Dec	Geometry and Physics Seminar, University of Sheffield, UK
Nov	QM Research Seminar, Centre for Quantum Mathematics, SDU, Denmark
Jul	29th Nordic Congress of Mathematicians with EMS, Aalborg, Denmark
Apr	Mathematics Seminar, IHES, France
Mar	Physical Mathematics Seminar, University of Geneva, Switzerland
2022, Nov	Workshop on Mathematics of Beyond All-Orders Phenomena, University of Cambridge, UK
2019, Nov	Seminar on Lie Groups and Moduli Spaces, University of Geneva, Switzerland
Aug	Conference on Representation Theory and Integrable Systems, ETH Zürich, Switzerland
2017, Dec	PADME Weekly Seminar, INFN – LNF, Italy

AWARDS AND SCHOLARSHIPS

2024 – 2027	Huawei Young Talents Program, IHES, France
2019 – 2020	Excellence Fellowship, NCCR SwissMAP, Switzerland
2019	Degree Prize for Distinction, St John's College, University of Oxford, UK
2018 – 2019	Torno Subito Scholarship, Department of Education, Research, and University, Lazio, Italy
2018	Best Student in Nuclear and Subnuclear Physics, Sapienza University of Rome and INFN, Italy
2017	Summer Student Scholarship, INFN, Italy
2016 – 2018	Excellence Program, Department of Physics, Sapienza University of Rome, Italy
2015 – 2018	Deserving Student Scholarship, Sapienza University of Rome, Italy

TEACHING EXPERIENCE

2024, Jan — Supervision of exercise sessions on Feynman Integrals and Number Theory
Winter School in Mathematical Physics, SwissMAP Research Station, Switzerland

2019, Oct — Lectures on Topological Surfaces
Master Class in Mathematical Physics, University of Geneva, Switzerland

2018, Mar – May — Lectures on Riemannian Geometry
Excellence Program in Physics, Sapienza University of Rome, Italy

SCIENTIFIC ACTIVITIES

COMMUNITY ENGAGEMENT

2026 – present — Co-organizer of the Monthly Women Scientists Lunch at IHES, France

PROFESSIONAL SERVICE

2026 – present — Referee for Communications in Number Theory and Physics
2025 — Co-organizer of a five-day workshop at Mathematisches Forschungsinstitut Oberwolfach, Germany
2020 – 2024 — Junior Member of the Scientific Council of the SwissMAP Research Station, Switzerland

RESEARCH NETWORKS

2024 – present — Associated Researcher, ERC SyG UNIVERSE+
2020 – 2024 — Doctoral Student, ERC SyG ReNewQuantum
2019 – 2024 — Excellence Fellow and Doctoral Student, NCCR SwissMAP

RESEARCH INTERNSHIPS

2020 — Research Internship in High-Energy Physics Phenomenology
CERN, Switzerland

2019 – 2020 — Master Class in Mathematical Physics
NCCR SwissMAP and University of Geneva, Switzerland

2017 — Research Internship in Experimental Particle Physics
INFN – LNF, Italy

SKILLS

Languages	Italian (native), English, French
Programming Languages	C, C++, Python, R
Data Analysis	MATLAB, ROOT, GNU plot
Symbolic Calculus Languages	Mathematica, SageMath, PARI/GP
Version Control Systems	Git
Simulation Software	Geant4